Biomarker-Guided Personalization of a Digital Substance-Use Mental Health Intervention

1. Introduction

Substance-use-related symptoms and impairment frequently co-occur with anxiety, depression, stress, impaired self-regulation, and disruptions in daily functioning. Although evidence-based digital mental health interventions may expand access to treatment, their benefits are often limited by variable engagement, inconsistent adherence, and differences in the amount and type of support that individuals require. A uniform intervention dose may be insufficient for some users and unnecessarily burdensome for others.

This concept note proposes a clinical trial to evaluate whether webcam-based eye-tracking and facial-expression signals can support safer and more precise personalization of an existing evidence-based digital mental health intervention for adults with clinically significant substance-use-related symptoms or functional impairment. The proposed work will build on the Brain-Gaze platform concept described by the Brain Rise Foundation, which integrates webcam-based eye tracking and facial-expression recognition to characterize responses to presented stimuli (Brain Rise Foundation, n.d.). However, the Brain-Gaze white paper presents a proposed and unvalidated concept for inferring neural or dopaminergic responses; it does not establish that webcam signals directly measure synaptic dopamine or that such signals improve clinical treatment.

Accordingly, the proposed study will use a staged design. First, it will assess the reliability, construct validity, and clinical validity of multimodal webcam-derived signals as treatment-relevant behavioral markers. Second, it will test whether biomarker-informed personalization of intervention intensity or support improves engagement, substance-use-related symptoms, and functioning compared with the same intervention delivered without biomarker personalization. The intervention will be delivered through an established digital health platform in partnership with the platform developer, consistent with the National Institute of Mental Health (NIMH) Laboratories to Optimize Digital Health NOFO (PAR-25-136).

The proposed research is responsive to the NOFO because it will: (1) leverage digital-platform functionality and multimodal behavioral data; (2) conduct research beyond a primary effectiveness trial; (3) test an empirically supported digital intervention in a population with clinically significant symptoms or impairment; (4) specify and evaluate a hypothesized mechanism of personalization; (5) address engagement, adherence, scalability, privacy, and safety; and (6) establish a foundation for regulatory consultation if the technology is later used to inform clinical decision-making.

2. Problem Statement

Digital mental health interventions can reduce access barriers, but treatment engagement and clinical response vary substantially across users. Existing platforms commonly provide the same content sequence, dose, and level of human or automated support to all participants. This approach may not adequately address differences in treatment-relevant response, cognitive and emotional engagement, craving-related reactivity, distress tolerance, or

capacity to sustain self-guided practice.

The proposed solution is not simply to develop another digital intervention. Rather, it is to determine whether existing platform functionality can identify valid, clinically meaningful response patterns and use those patterns to optimize delivery of an evidence-based intervention. The central scientific problem is that the proposed webcam-based signals have not yet been adequately validated for this purpose. The Brain-Gaze white paper hypothesizes that eye tracking and facial-expression recognition may provide information about physiological or emotional responses to stimuli and may eventually support models of dopamine-related response (Brain Rise Foundation, n.d.). These claims remain preliminary. Facial expression and gaze behavior are not direct measures of synaptic dopamine, and observed signals may be influenced by lighting, camera quality, culture, disability, medication, fatigue, mood, privacy-related behavior, and other factors.

Three related gaps must therefore be addressed:

1. **Measurement gap:** The reliability and validity of webcam-derived eye-tracking and facial-expression features for treatment-relevant behavioral states are not established in the proposed clinical population.
2. **Personalization gap:** It is unknown whether such signals can identify individuals who benefit from different intervention doses, sequences, or levels of support.
3. **Clinical utility and safety gap:** It is unknown whether biomarker-guided personalization produces clinically meaningful improvement without increasing burden, inequity, privacy risk, false alerts, or unsafe treatment recommendations.

The study will address the following research questions:

1. Are webcam-based eye-tracking and facial-expression signals sufficiently reliable across repeated assessments and common real-world recording conditions?
2. Do these signals demonstrate convergent and predictive validity with validated measures of craving, affective response, engagement, symptoms, and functioning?
3. Can multimodal signals improve prediction of near-term intervention engagement or clinical response beyond baseline clinical and demographic information?
4. Does biomarker-guided personalization of intervention dose or support improve engagement and clinical outcomes compared with the same intervention delivered without biomarker personalization?
5. Does target engagement—defined as improved matching of intervention intensity or support to treatment-relevant response—mediate or explain the effect of personalization?
6. Are the approach's performance and effects consistent across sex, age, race and ethnicity, disability status, socioeconomic circumstances, and device or recording conditions?

3. Objectives

The long-term goal is to develop scalable, safe, and equitable methods for matching digital mental health treatment intensity and support to the needs of individuals with substance-use-related symptoms and impairment.

The objective of this project is to validate a multimodal webcam-based personalization approach and determine whether it improves delivery and outcomes of an existing evidence-based digital intervention. The study will pursue the following measurable objectives:

1. **Characterize measurement reliability.**

Quantify the test-retest reliability, internal consistency where applicable, missingness, signal quality, and sensitivity of candidate eye-tracking and facial-expression features across repeated assessments and varied webcam conditions.

2. **Evaluate clinical and behavioral validity.**
   Estimate associations between candidate features and validated measures of craving, affective response, treatment engagement, substance-use-related symptoms, and functional impairment. The study will distinguish associations with observable behavioral and clinical constructs from any claims about direct neural or dopaminergic measurement.

3. **Develop and internally validate a treatment-matching algorithm.**
   Determine whether multimodal features add predictive value beyond standard baseline information, including symptom severity, prior treatment exposure, motivation, and early platform engagement. The algorithm will be locked before confirmatory outcome analyses.

4. **Test biomarker-guided personalization in a randomized clinical trial.**
   Compare the existing evidence-based digital intervention delivered with biomarker-informed dosing or support against the same intervention delivered using a prespecified non-personalized protocol. Both groups will receive an active intervention; the study will not use a non-active comparator.

5. **Evaluate clinical, functional, and implementation outcomes.**
   Assess between-group differences in substance-use-related symptoms, functional impairment, sustained skill use, treatment engagement, intervention completion, user burden, satisfaction, health-services use where feasible, and safety events.

6. **Test the hypothesized mechanism of personalization.**
   Determine whether biomarker-guided care improves outcomes by increasing the fit between treatment intensity or support and treatment-relevant response, as reflected by target engagement, intervention exposure, sustained skill practice, and proximal changes in craving or affective regulation.

7. **Assess equity, privacy, safety, and scalability.**
   Examine model performance and treatment effects across relevant demographic and access-related subgroups; quantify data-quality disparities; evaluate participant concerns about facial and gaze data; and identify implementation requirements for routine digital care.

Expected outcomes include: (a) a validated or qualified set of webcam-derived behavioral features, or a clear determination that the signals are not sufficiently reliable for personalization; (b) a preregistered treatment-matching algorithm and analysis framework; (c) an estimate of the incremental clinical and engagement benefit of personalization; and (d) evidence to guide future deployment, platform refinement, data-sharing, and potential FDA or other regulatory consultation.

4. Preliminary Literature Review

The Brain-Gaze white paper describes a proposed platform that combines webcam-based eye tracking and facial-expression recognition for real-time analysis of responses to digital stimuli (Brain Rise Foundation, n.d.). It identifies consumer neuroscience methods such as eye tracking, pupillometry, galvanic skin response, and facial-expression recognition as potentially useful for measuring responses that may not be captured by self-report. The paper

further proposes that a laboratory stimulus paradigm associated with rapid dopamine changes could be used to generate training data for machine-learning models. These proposals provide a conceptual starting point for the present study but do not establish clinical validity, diagnostic accuracy, treatment utility, or direct measurement of synaptic dopamine through a webcam.

The white paper's account of addiction is grounded in established neuroscience concerning reward learning, cue reactivity, motivation, and changes in cognitive control. Contemporary addiction models describe substance use as involving interactions among reward circuitry, conditioned cues, stress systems, learning, craving, and impaired control (Koob and Volkow 2016; Volkow, Michaelides, and Baler 2019). Research on drug craving similarly supports the importance of cue-related responses and the translation of neurobehavioral mechanisms into treatment research (Ekhtiari et al. 2016). These findings support investigating treatment-relevant behavioral responses, but they do not demonstrate that a single webcam-derived signal can serve as a direct proxy for dopamine or a clinically actionable biomarker.

The white paper also cites work pairing invasive fast-scan cyclic voltammetry with machine learning to detect rapid dopamine changes during neurosurgical procedures and proposes adapting laboratory reward paradigms for noninvasive computer-vision research (Brain Rise Foundation, n.d.). Such work may inform stimulus design and mechanistic hypotheses, but invasive measurements obtained in specialized clinical settings cannot be assumed to transfer directly to webcam-based recordings in community or home environments. The proposed MSRU study will therefore treat the webcam outputs as candidate behavioral markers requiring independent validation.

Facial-expression recognition also presents important methodological challenges. Performance can vary by context, lighting, camera position, skin tone, facial mobility, disability, cultural display rules, and the distinction between an expressed emotion and a person's underlying affective state. The open-source FeelPix project is described in the white paper as a resource for validating facial-expression recognition tools (La Monica et al. 2023). In the proposed study, such tools may support technical benchmarking, but clinical validity will require comparison with validated self-report, behavioral, and clinical measures rather than reliance on expression labels alone.

The NIMH NOFO calls for digital mental health studies that leverage established platforms, test interventions in individuals with clinically significant symptoms or impairment, use active comparators where appropriate, and go beyond a primary effectiveness trial by capitalizing on platform-specific data and functionality (National Institute of Mental Health 2024). It also requires explicit specification and assessment of intervention targets and mechanisms. The present concept addresses these requirements by combining measurement validation, algorithm development, and a randomized clinical trial of personalization. The presumed mechanism is not "dopamine measurement" itself. The proposed mechanism is improved matching of intervention dose or support to treatment-relevant response, leading to stronger engagement, more sustained skill practice, and improved symptoms and functioning.

Important limitations remain. The supporting white paper does not report a completed clinical validation study, reliability estimates, sample characteristics, treatment effect sizes, or evidence that the prototype has an established active user base. Before application development, MSRU and its prospective platform partner will need to document platform maturity, existing user volume, intervention evidence, technical access, data governance, and recruitment feasibility. These are essential because the NOFO expects use of a well-established digital health platform with sufficient users and infrastructure to support appropriately powered, deployment-focused research.

5. Methodology

Study design

The proposed project will use a sequential mixed-methods and randomized clinical-trial design over a maximum four-year period, subject to feasibility, platform readiness, investigator input, and NIMH consultation.

The study will include three linked components:

1. **Technical and measurement validation:** repeated webcam sessions and standardized digital stimuli to assess signal quality, reliability, and convergent validity.
2. **Algorithm development and treatment matching:** analysis of baseline and early-use data to identify whether multimodal signals improve prediction of engagement or clinical response.
3. **Randomized clinical trial:** comparison of biomarker-guided personalization with the same evidence-based digital intervention delivered according to a fixed, non-personalized protocol.

The selected intervention must already have empirical support for reducing clinically significant mental health symptoms or functional impairment and must be available through an established digital platform. The final intervention, platform developer, intervention dose, and level of human or automated support will be specified after feasibility review and consultation with NIMH.

Participants and setting

Participants will be adults who meet prespecified criteria for clinically significant substance-use-related symptoms or impairment at baseline. Screening will assess substance-use symptoms, mental health symptoms, functional impairment, current treatment, medication status, technology access, and safety factors. The study may include individuals with co-occurring anxiety, depression, trauma-related symptoms, or other mental health concerns when consistent with the selected intervention and safety plan.

Potential exclusion criteria may include circumstances in which remote digital participation is unsafe or scientifically inappropriate, such as acute intoxication requiring emergency care, inability to provide informed consent, or clinical instability requiring a higher level of treatment. These criteria will be finalized with clinical investigators and the IRB.

Recruitment will occur through the established platform, collaborating treatment and community organizations, and other approved channels. Recruitment and retention plans will intentionally address populations experiencing disparities in access to mental health care. The final sample size will be determined through a formal power analysis based on the selected primary clinical outcome, the expected incremental effect of personalization, attrition, clustering or repeated-measures structure, and the number of parameters in the treatment-matching algorithm.

Intervention and comparator

All participants will receive the same evidence-based digital intervention. The trial will compare:

- **Biomarker-guided personalization:** webcam-derived signals, together with

prespecified clinical and engagement information, will inform intervention dose, sequence, reminders, or level of support according to an intervention policy defined before the confirmatory trial; and
- **Non-personalized active intervention:** the same intervention content and platform will be delivered according to a fixed protocol that does not use biomarker-based personalization.

The personalization policy will not independently diagnose substance-use disorder, determine medication dosing, make emergency decisions, or replace clinician judgment. During the trial, the algorithm will be restricted to low-risk delivery decisions, such as the timing of reminders, selection among already approved modules, pacing, or offering additional support. Any escalation for suicide risk, severe withdrawal, overdose risk, or acute psychiatric deterioration will follow a separate clinical safety protocol.

Measurement validation

Candidate features will include webcam-based gaze position or attention-related measures, blink or fixation characteristics where technically feasible, facial movement or expression features, and multimodal temporal patterns. The study will record signal quality indicators and contextual variables such as lighting, camera type, background, device, and interruptions.

Validation measures may include:

- validated substance-use symptom and craving measures;
- measures of depression, anxiety, stress, and emotion regulation;
- behavioral indicators of attention, persistence, and task completion;
- self-reported affective response to standardized stimuli;
- platform engagement and intervention exposure;
- functional impairment and quality of life; and
- clinician- or researcher-administered assessments where feasible.

Reliability analyses will estimate test-retest reliability, missingness, signal degradation, and agreement across recording conditions. Construct-validity analyses will examine prespecified associations with behavioral and clinical measures. Predictive-validity analyses will assess whether early signals predict subsequent engagement, skill use, symptoms, or impairment. Analyses will explicitly test whether candidate features add information beyond conventional clinical variables.

The study will not label webcam outputs as direct measures of synaptic dopamine unless independent evidence supports that interpretation. If the data support only behavioral or affective interpretation, the resulting algorithm and dissemination materials will use appropriately limited terminology.

Randomization and outcomes

Following baseline assessment and completion of the validation procedures required for safe algorithm use, eligible participants will be randomized to biomarker-guided personalization or non-personalized active intervention. Randomization will be stratified as appropriate by baseline symptom severity, substance category or treatment-relevant subgroup, and other variables identified before enrollment.

The primary clinical outcome will be a validated measure of substance-use-related symptoms or impairment, selected in consultation with clinical experts and the platform partner.

Primary and secondary outcomes will be assessed at baseline, during treatment, immediately after the intervention period, and during follow-up. The observation period will be long enough to evaluate initial symptom change, sustained engagement, skill use, and maintenance of benefit; the final schedule will depend on the intervention's intended treatment window.

Secondary outcomes will include:

- treatment engagement and retention;
- intervention completion and dose received;
- sustained use of intervention skills;
- craving and affective response;
- functioning and quality of life;
- service utilization and treatment continuity where available;
- user burden, acceptability, and satisfaction;
- privacy concerns and willingness to continue webcam-enabled care;
- adverse events and safety events; and
- subgroup-specific performance and outcomes.

The study will define clinically meaningful improvement before unblinded outcome analysis. Analyses will use intention-to-treat principles, appropriate methods for repeated measures and missing data, and sensitivity analyses for adherence and protocol deviations.

Mechanism and mediation analysis

The conceptual model is:

Webcam-derived behavioral signals → improved treatment matching → increased engagement and skill practice → reduced symptoms and functional impairment.

Target engagement will be assessed through:

1. whether the algorithm produces different and clinically interpretable delivery decisions across participants;
2. whether those decisions improve intervention exposure and completion;
3. whether participants receiving matched support demonstrate greater sustained skill practice;
4. whether proximal changes in craving, affective response, or self-regulation occur before clinical improvement; and
5. whether changes in these proximal targets statistically account for the association between personalization and clinical outcomes.

Mediation analyses will be exploratory unless the sample size and measurement schedule support confirmatory inference. The analysis plan will distinguish algorithm performance, treatment-process effects, and clinical outcomes. It will also evaluate whether personalization benefits particular subgroups or produces unequal error rates.

Data analysis and algorithm governance

The algorithm-development dataset will be separated from the confirmatory trial dataset to reduce overfitting. Candidate models will be compared with baseline-only models, and performance will be evaluated using prespecified metrics appropriate to prediction and classification. The final algorithm will be locked before confirmatory outcome analyses.

The study will monitor calibration, missing data, subgroup performance, and changes in platform or camera technology. Software updates that affect signal extraction or intervention delivery will be version-controlled, documented, and evaluated before implementation. Any post-lock changes will be treated as protocol amendments and analyzed separately when appropriate.

Privacy, safety, and regulatory considerations

Facial and gaze data are sensitive biometric and behavioral information. Data collection will be minimized to the information necessary for the research questions. The study will consider on-device processing, encryption, access controls, separation of identifying information from research data, retention limits, and participant options regarding image storage. Participants will be informed that webcam signals are experimental and will not be represented as a definitive measure of dopamine, diagnosis, relapse risk, or personal character.

A clinical safety plan will address suicidality, severe substance-use complications, overdose risk, withdrawal concerns, acute psychiatric deterioration, and technical failures that could interrupt support. The digital intervention will provide clear crisis and emergency instructions, and clinical staff will follow prespecified response procedures.

Because the technology could eventually influence clinical care, MSRU and the platform developer should seek early consultation regarding whether the software constitutes a medical device or clinical decision-support function and whether FDA, institutional, state, or other regulatory requirements apply. The research protocol will be reviewed by the appropriate IRB, and data and safety monitoring will follow NIH and NIMH requirements for clinical trials.

Stakeholder and deployment engagement

The project will involve people with lived experience of substance-use-related symptoms, treatment providers, community organizations, digital-platform developers, privacy and security experts, and relevant health-system or payer stakeholders. Their input will inform:

- acceptability of webcam-based data collection;
- intervention burden and support thresholds;
- safety messaging and escalation procedures;
- privacy protections and consent language;
- feasibility in routine service settings;
- interpretation of clinically meaningful improvement; and
- dissemination and implementation requirements.

The platform partner will be expected to document current active-user volume, infrastructure for remote consent and assessment, data-quality procedures, intervention fidelity, and the feasibility of recruiting and retaining a sufficiently diverse sample. If these requirements cannot be verified, the project will be redesigned or submitted to a funding opportunity better suited to platform development or early feasibility work.

Data management and sharing

The study will use standardized mental health common data elements where applicable and will prepare deidentified datasets, metadata, analytic documentation, and code for sharing consistent with NIH and NIMH requirements. The team will develop a Data Management and

Sharing Plan and address submission to the NIMH Data Archive, including participant identifiers required for approved data integration, data contribution costs, quality control, and restrictions on sharing sensitive facial or video data.

Because raw facial recordings may create re-identification risks, the investigators will determine whether derived features, quality metrics, synthetic data, or controlled-access data can satisfy scientific and policy requirements without exposing participants to unnecessary privacy risk.

Proposed project schedule

Period	Major activities
Year 1	Confirm platform and intervention eligibility; establish developer and clinical partnerships; stakeholder consultation; protocol development; measurement and privacy review; pilot technical testing; IRB and regulatory consultation; finalize power analysis and analysis plan.
Year 2	Conduct measurement-validation study; assess reliability, signal quality, and clinical validity; refine and lock candidate features; develop and internally validate treatment-matching algorithm; establish recruitment and safety-monitoring procedures.
Year 3	Conduct randomized clinical trial enrollment and intervention delivery; monitor fidelity, engagement, adverse events, data quality, and subgroup performance; implement approved platform refinements.
Year 4	Complete follow-up; conduct primary, secondary, mechanism, equity, and implementation analyses; prepare NDA and other data-sharing submissions; disseminate findings; determine whether further clinical, implementation, or regulatory studies are warranted.

References

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